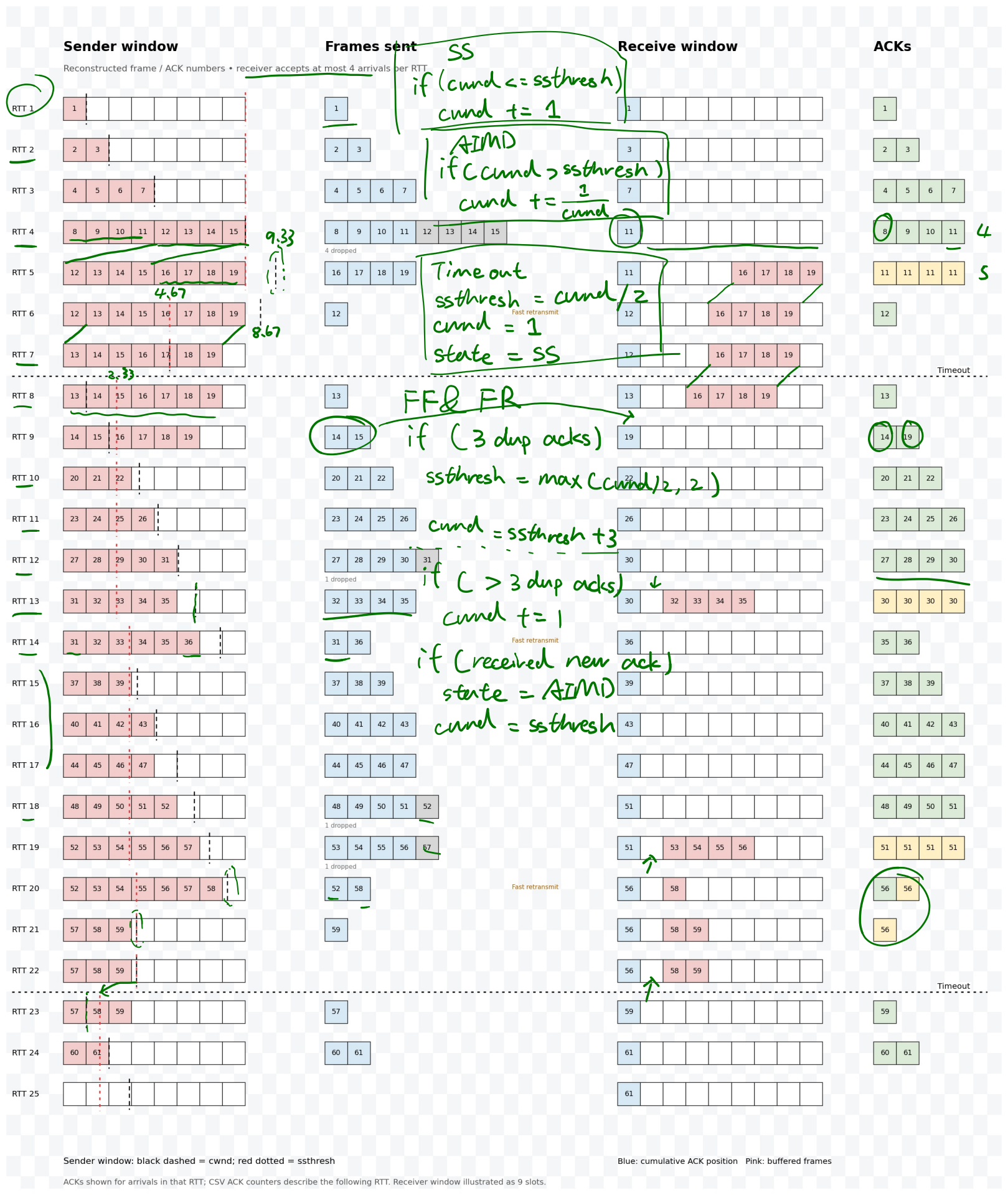

Sender window
Frames sent
Receive window
ACKs
Reconstructed frame / ACK numbers • receiver accepts at most 4 arrivals per RTT
RTT 1
RTT 2
RTT 3
RTT 4
RTT 5
RTT 6
RTT 7
RTT 8
RTT 9
RTT 10
RTT 11
RTT 12
RTT 13
RTT 14
RTT 15
RTT 16
RTT 17
RTT 18
RTT 19
RTT 20
RTT 21
RTT 22
RTT 23
RTT 24
RTT 25
9.33
4.67
8.67
2.33
4 dropped
1 dropped
1 dropped
1 dropped
Fast retransmit
Fast retransmit
Fast retransmit
Timeout
Timeout
SS
if (cwnd <= ssthresh)
cwnd += 1
AIMD
if (cwnd > ssthresh)
cwnd += 1/cwnd
Timeout
ssthresh = cwnd/2
cwnd = 1
state = SS
FF & FR
if (3 dup acks)
ssthresh = max(cwnd/2, 2)
cwnd = ssthresh + 3
if (> 3 dup acks)
cwnd += 1
if (received new ack)
state = AIMD
cwnd = ssthresh
Sender window: black dashed = cwnd; red dotted = ssthresh
Blue: cumulative ACK position
Pink: buffered frames
ACKs shown for arrivals in that RTT; CSV ACK counters describe the following RTT. Receiver window illustrated as 9 slots.